

FULL STACK TECHNOLOGIES

LAB 5

AKSHAYA.R

2024103532

TOPICS COVERED:

- Express.js concepts - Server, file handling ,url, use,static, loading data from json and writing data into json, downloading file.

ADD.HTML

```
<? add.html X
labc > <? add.html > style
1 <div><form method="post">
2 Enter Title <br><input type="text" name="title"><br>
3 Enter Author<br> <input type="text" name="author"><br><br>
4 <button type="button" onclick="addNewAuthor()">Add More Authors</button>
5 <input type="submit" value="Add Book">
6 </form>
7 </div>
8 <script>
9     function addNewAuthor(){
10         const a=document.querySelector('button');
11         a.insertAdjacentHTML('beforebegin','<input type="text" name="author"><br><br>');
12     }
13 </script>
14 <style>
15     div{
16         text-align:center;
17         padding-top:50px;
18     }
19 </style>
```

BOOK.HTML

```
<? add.html <? book.html X
labc > <? book.html > table
1 <table border="1"><tr><th>Title</th><th>Author</th><th>Download Link</th></tr><tr><td>HPP</td><td>DD,BB,CC</td><td><a href="./download/1.txt" download="HPP.txt">Download</a></td></tr>
2 <tr><td>nblk</td><td>jfl,nbkl,mknbl</td><td><a href="./download/2.txt" download="nblk.txt">Download</a></td></tr>
3 <tr><td>HP</td><td>AA,BB,CC</td><td><a href="./download/3.txt" download="HP.txt">Download</a></td></tr>
4 <tr><td>HPP</td><td>DD,BB,CC</td><td><a href="./download/4.txt" download="HPP.txt">Download</a></td></tr>
5 <tr><td>mjkbv</td><td>jbv.f,fd,m,kbcn</td><td><a href="./download/5.txt" download="mjkbv.txt">Download</a></td></tr>
6 <tr><td>HP</td><td>AA,BB,CC</td><td><a href="./download/6.txt" download="HP.txt">Download</a></td></tr>
7 <tr><td>HPP</td><td>DD,BB,CC</td><td><a href="./download/7.txt" download="HPP.txt">Download</a></td></tr>
8 <tr><td>HP</td><td>AA,BB,CC</td><td><a href="./download/8.txt" download="HP.txt">Download</a></td></tr>
9 <tr><td>HPP</td><td>DD,BB,CC</td><td><a href="./download/9.txt" download="HPP.txt">Download</a></td></tr>
10 <tr><td>nxcbm</td><td>mcx,,m, zbx,,kj cdvk</td><td><a href="./download/10.txt" download="nxcbm.txt">Download</a></td></tr>
11 </table>
```

LIBRARY.JS

labc > labexp > JS library.js > getbookauthor

```
1  const a=require('fs')
2  let books=[]
3  let data=a.readFileSync('./datas.json')
4  books=JSON.parse(data.toString())
5  var i=books.length
6  function addbook(title,author){
7    books[i]={title:title,author:author,filename:`${i}.txt`}
8    a.writeFile(`./books/${i}.txt`,`${title},${author}`.repeat(1000000),()=>{ })
9    i++
10   a.writeFileSync("datas.json",JSON.stringify(books),()=>{ })
11 }
12 function getbookauthor(title)
13 {
14   for(let j=0;j<i;j++)
15   {
16     if(books[j].title==title)
17     {
18       return books[j].author;
19     }
20   }
21 }
22 function getbook(title)
23 {
24   for(let j=0;j<i;j++)
25   {
26     if(books[j].title==title)
27     {
28       return books[j].filename;
29     }
30   }
31 }
32 function getbooktitle(author)
33 {
34   var authors=[]
35   for(let j=0;j<i;j++)
36   {
```

```
function getbooktitle(author)
{
  var authors=[]
  for(let j=0;j<i;j++)
  {
    for(let k=0;k<books[j].author.length;k++)
    {
      if(books[j].author[k]==author)
      {
        authors.push(books[j].title)
      }
    }
  }
  return authors
}
function getbookna(){
  var na=[]
  for(let j=0;j<i;j++)
  {
    na[j]={title:books[j].title,author:books[j].author}
  }
  return na;
}
function getTitle(filename)
{
  for(let j=0;j<i;j++)
  {
    if(books[j].filename==filename)
    {
      return books[j].title;
    }
  }
}

module.exports={addbook,getTitle}
addbook("hdgf",["yk","ll"])
```

EXP.JS

labc > labexp > JS exp.js > a.get('/download/tit') callback

```
2  const e=require('express')
3  const d=require('./library.js')
4  const c=require('fs')
5  const a=e()
6  a.listen(3000)
7  var count=0
8  var downloadCount=0
9
10 const downloadInfo=c.readFileSync("./downloadinfo.json")
11 var downloadData=JSON.parse(downloadInfo.toString())
12 var totalDownload=downloadData.totalDownloads
13 a.get('/book',(req,res)=>
14 {
15   var data=c.readFileSync("./datas.json")
16   let books=JSON.parse(data.toString())
17   c.writeFileSync("./book.html","<table border='1'><tr><th>Title</th><th>author</th><th>download link</th></tr>")
18   for(let j=0;j<books.length;j++)
19   {
20     c.appendFileSync("./book.html`,`<tr><td>${books[j].title}</td><td>${books[j].author}</td><td><a href='/download/${books[j].filename}'>Download</a></td></tr>")
21   }
22   c.appendFileSync("./book.html","</table>")
23   resp.sendFile("book.html",{root:__dirname})
24 })
25 a.use((req,res,next)=>{
26   count++
27   next()
28   console.log(count)
29 })
30 a.get('/add',(req,res)=>{
31   resp.sendFile("add.html",{root:__dirname})
32 })
33 a.use(e.static('abc'))
34 a.use(e.urlencoded({extended:true}))
35 a.post('/add',(req,res)=>{
36   let title=req.body.title
37   let author=req.body.author
```

```
add.html JS library.js JS exp.js X
labc > labexp > JS exp.js > a.get('/download/tit') callback
25 a.use((req,res,next)=>{
28 | console.log(count)
29 | })
30 a.get('/add',(req,res)=>{
31 | resp.sendFile("add.html",{root:__dirname})
32 | })
33 a.use(e.static('abc'))
34 a.use(e.urlencoded({extended:true}))
35 a.post('/add',(req,res)=>{
36 | let title=req.body.title
37 | let author=req.body.author
38 | d.addbook(title,author)
39 | resp.send("<h3>Book added Successfully</h3><a href='./book'>View Books</a>")
40 | })
41
42 a.get('/download/:tit',(req,res)=>
43 | {
44 |   const title=req.params.tit;
45 |   console.log(title)
46 |   resp.download(`/books/${title}`,d.getTitle(title},{root:__dirname})
47 |   totalDownloads++;
48 |   downloadCount++;
49 |   c.writeFileSync("./downloadinfo.json",JSON.stringify({totalDownloads:totalDownloads}))
50 | })
51 setInterval(()=>{
52 |   console.log("Downloads :"+downloadCount);
53 | },300000)
54 setInterval(()=>{
55 |   console.log("Downloads in last 30 minutes:"+downloadCount);
56 |   console.log("Total Downloads:"+totalDownloads);
57 |   downloadCount=0;
58 | },1800000)
```

DATA.JSON

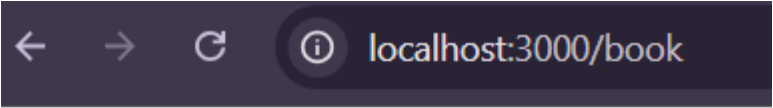
```
add.html {} data.json X
bc > {} data.json > ...
1 [{"title":"HP","author":["AA","BB","CC"],"filename":"0.txt"},{"title":"HPP","author":["DD","BB","CC"],"filename":"1.txt"},{"title":"nblk","author":{
```

DOWNLOADINFO.JSON

```
add.html {} downloadinfo.json X
labc > {} downloadinfo.json > ...
1 [{"totalDownloads":2}]
```

OUTPUT:

localhost:3000/book



Title	author	download link
hdgf	yk,ll	Download
mfdn	c m,mdkvbj	Download
hdgf	yk,ll	Download
hdgf	yk,ll	Download
hdgf	yk,ll	Download
hdgf	yk,ll	Download
hdgf	yk,ll	Download

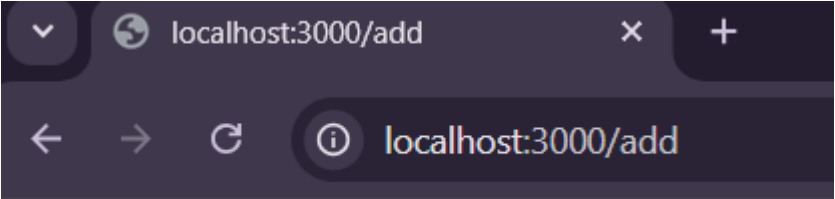
localhost:3000/add



Enter Title

Enter Author

On clicking add book,it redirects to book page



Book added Successfully

[View Books](#)

On clicking download ,file gets downloaded

A screenshot of a web browser window. The address bar shows 'localhost:3000/book'. The page displays a table with three columns: 'Title', 'author', and 'download link'. The table contains eight rows of data. A download notification is visible in the bottom right corner, showing a file named 'hdgfg (3)' with a size of 9.5 MB and a status of 'Done'.

DOWNLOADED FILE:

[illegible]